

Layered T Full Adder using Quantum-dot Cellular Automata

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Abstract— Fast changing world has driven technology towards a milestone where emerging technology like quantum-dot cellular automata (QCA) excels in terms of ultra-high packing density and extremely low power consumption. Quantum Cellular Automata has several deduction methodologies like Majority Voter, Universal QCA logic, FNZ logic & And-Or-Invert (AOI) logic none of which explores universal NAND-NOR based design in Boolean reduction techniques. This work proposes Layered T full adder with its basic primitive which works on the basis of universal NAND and NOR logic. Layered T gate is verified taking Coulomb's law as the physics behind it. Layered T Gate is also used to implement full adder as primitive in processor based design which gives the best result in terms of cell requirement and area compared to the latest design.

Keywords—Universal NAND/NOR based Logic, Potential calculation in QCA, Full adder

I. INTRODUCTION

With the advent of nanotechnology, size and power consumption of electronic devices have become key concerns. When the efficiency comes with devices having size in nano-scale, CMOS technology is found inefficient because of its non-linear characteristics below sub nanometer regime [1]. Challenges of recent Si FinFET left Quantum Cellular Automata in a better position where nonlinear effects are eluded [2]. Quantum-dot cellular automaton projects a viewpoint where information flows through columbic interaction [3] [4]. Quantum cell of 18nmX18nm bounds two excess electrons at the furthest positions in either direction in such a way that they can achieve minimum repulsive force between them [5].

Quantum-dot Cellular Automata focuses on Majority Logic [5], Universal QCA Logic Gate [6], FNZ gate [7] & AOI Gate [8]. But lack of practices still exist which focus on universal NAND/NOR based logic. This work familiarizes a new layer base design which imposes

universality and requires five QCA cells to work. Working of proposed layer base primitive is validated with the instantiation of full adder and compared with previous full adder designs [3] [6] [16-17] [21-23] in terms of area and cell count.

This work starts with brief background of QCA in section II where the succeeding section introduces the details of proposed layered T gate. Section III considers not only its structures but also points on optimized configuration. Section IV focuses on scalability of this gate in full adder design. This section compares the adder with other previously proposed designs. Section V observes the results of proposed schematic and lastly section VI concludes the entire work.

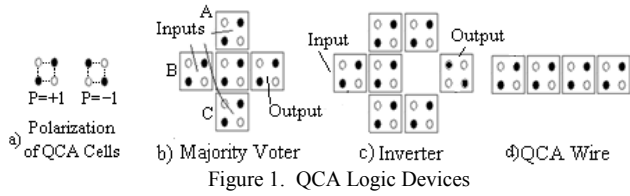
II. BACKGROUND OF QCA

A. QCA Logic Devices

Electrostatic repulsion forces two excess electrons to arrange at the furthest corners within a QCA cell. As a result, two possible configurations can be built with polarization “+1” (logic 1) or “-1” (logic 0) as shown in figure 1a. Different logical structures can be instantiated if these QCA cells are arranged accordingly, given as a) Majority Voter, b) QCA Inverter c) QCA Wire shown by figure 1b, figure 1c & figure 1d respectively. Majority gate has three inputs A, B & C with the Boolean function $M(A,B,C)=AB+BC+CA$ as shown in figure 1b.

B. QCA Clocking

In QCA, the charge flows from input port to output port upon the arrival of clock pulse. This clocking scheme has got four phases with 90° phase difference between each other. Namely “switch”, where logical computation is evaluated, “hold” phase retain the charge within a cell, “release” passes the information to the next cell and lastly “relax” phase grounds the system [9].



One cell or preferably a group of cells works under same clock phase which forms level of a circuit. The intermediary results from a specific level is evaluated by means of the given clock phase. Complex QCA circuits are divided into group of cells work under the same clock phase termed as clock zone [10].

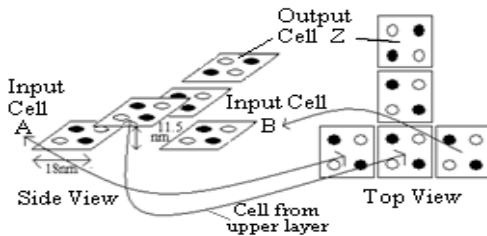
III. LAYERED T-METHODOLOGY & ITS VERIFICATION

A. Proposed Layered T-Gate

Five input cells are organized as shown in figure 2 and figure 3 which consider both top view and side view of the proposed gate. Top view of the gate resembles the letter 'T' and side view needs a layer for fixing the upper cell. Hence the name Layered T Gate. Two input cells A and B arrange themselves in main cell layer with output cell Z whereas fixed polarization cell lies in upper layer, 11.5 nm above the main cell layer. Polarization "+1" of upper layer cell generates the NAND output at Z but "-1" produces NOR output. To illustrate the working of this gate, following case studies are considered where fixed polarization given is +1.

Case 1: When the inputs are identical (either "+1" or "-1"), polarization from two inputs dominate over the fixed polarized cell which makes strong inversion of input polarization. It means if both the inputs are "+1", the output becomes "-1" (logic 0) and vice versa.

Case 2: If the polarizations of the inputs are different, fixed cell polarization dominates and it influences



the output cell in favor of its own polarity. It shows that for Layered T gate output will be at logic 1 if inputs are different. Entire case studies show that if fixed polarization given to upper layer cell is "+1", output looks like NAND gate as shown in figure 4a. Similarly NOR logic can be applicable if upper layer cell polarization becomes "-1". Output show the resemblance NOR output as shown in figure 4b.

The following expressions show the functioning of the gate:

$$T_L^+(A,B)=\overline{AB} \quad (1)$$

$$T_L^-(A,B)=\overline{A+B} \quad (2)$$

Here the letter 'T' signifies T-gate and subscript 'L' implies the use of layer in its layout.

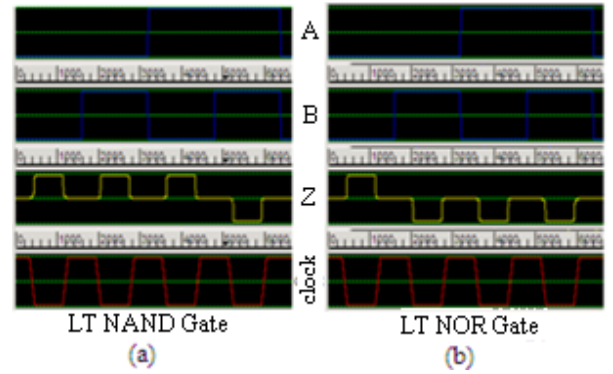
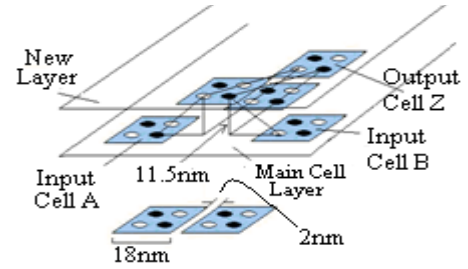


Table I Parameter used in potential calculation

Parameter	Definition
q	Electronic charge
U	Potential of electron at a specific distance
k	Proportionality constant
r	distance between two specified electrons
U_{T+1}	Total potential for polarization +1
U_{T-1}	Total potential for polarization -1
U_{T+1x}	Potential for polarization +1 for electron x in specific configuration
U_{T+1y}	Potential for polarization +1 for electron y in specific configuration
U_{T-1x}	Potential for polarization -1 for electron x in specific configuration
U_{T-1y}	Potential for polarization -1 for electron y in specific configuration

B. Potential Stability- Layered T Gate

Device engineer and manufacturer could make arguments that if any one removes the cell from layout given in figure 5; it gives the output which might have the same result to that of figure 6. This ambiguity can be cleared in terms of potential stability consideration. To make a configuration stable, QCA cells should be arranged in such a manner that the electrostatic interaction or the potential between them are minimal [11]. Considering potential expression [12] as shown in equation (3):

$$U = \frac{kq_1q_2}{r} \quad (3)$$

Table II Potential calculation of Configuration 1 of figure 5

Potential for P=+1 of configuration 1						Potential for P=-1 of configuration 1					
Sl. No.	Distance from electron x	Energy (in J)	Sl. No.	Distance from electron y	Energy (in J)	Sl. No.	Distance from electron y	Energy (in J)	Sl. No.	Distance from electron y	Energy (in J)
1.	to $e_1=44.72\text{nm}$	0.515×10^{-20}	9.	to $e_1=69.34\text{nm}$	0.3329×10^{-20}	1.	to $e_1=61.35\text{nm}$	0.376×10^{-20}	9.	to $e_1=55.17\text{nm}$	0.418×10^{-20}
2.	to $e_2=22.09\text{nm}$	1.045×10^{-20}	10.	to $e_2=44.72\text{nm}$	0.516×10^{-20}	2.	to $e_2=40.04\text{nm}$	0.576×10^{-20}	10.	to $e_2=40.04\text{nm}$	0.576×10^{-20}
3.	to $e_3=41.62\text{nm}$	0.554×10^{-20}	11.	to $e_3=61.8\text{nm}$	0.37×10^{-20}	3.	to $e_3=59.12\text{nm}$	0.39×10^{-20}	11.	to $e_3=59.12\text{nm}$	0.39×10^{-20}
4.	to $e_4=30.66\text{nm}$	0.753×10^{-20}	12.	to $e_4=41.62\text{nm}$	0.554×10^{-20}	4.	to $e_4=45.34\text{nm}$	0.509×10^{-20}	12.	to $e_4=24.84\text{nm}$	0.929×10^{-20}
5.	to $e_5=29.73\text{nm}$	0.776×10^{-20}	13.	to $e_5=40.049\text{nm}$	0.5765×10^{-20}	5.	to $e_5=61.35\text{nm}$	0.376×10^{-20}	13.	to $e_5=40.049\text{nm}$	0.5765×10^{-20}
6.	to $e_6=55.17\text{nm}$	0.433×10^{-20}	14.	to $e_6=61.35\text{nm}$	0.376×10^{-20}	6.	to $e_6=29.73\text{nm}$	0.776×10^{-20}	14.	to $e_6=29.732\text{nm}$	0.776×10^{-20}
7.	to $e_7=20\text{nm}$	1.15×10^{-20}	15.	to $e_7=42.047\text{nm}$	0.549×10^{-20}	7.	to $e_7=20\text{nm}$	1.15×10^{-20}	15.	to $e_7=18\text{nm}$	1.28×10^{-20}
8.	to $e_8=18.11\text{nm}$	1.27×10^{-20}	16.	to $e_8=20\text{nm}$	1.15×10^{-20}	8.	to $e_8=42.04\text{nm}$	0.54×10^{-20}	16.	to $e_8=18.11\text{nm}$	1.274×10^{-20}
U_{T+1x}		6.5×10^{-20}	U_{T+1y}		4.42×10^{-20}	U_{T-1x}		4.70×10^{-20}	U_{T-1y}		6.22×10^{-20}

where q_1 and q_2 are electronic charges and r is the distance between them. Five cell Layered T Gate is shown in configuration 1, mapped to figure 5 whereas four cell Layered T Gate comes in configuration 2, mapped to figure 6.

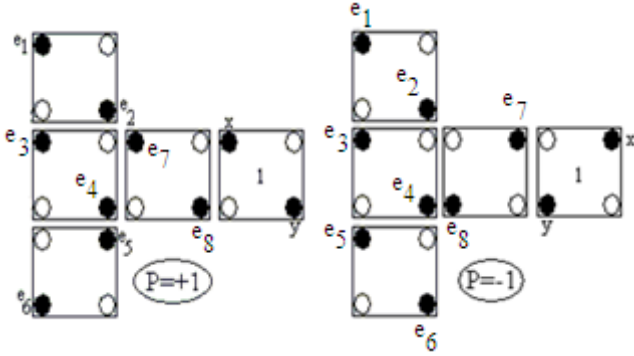


Figure 5. Configuration 1 of Layered T gate (five cell structure)

Calculation of potential for electron x in configuration 1 is given in accordance with table I:

Polarization $P = +1$: The distance between electron x and electron e_1 is calculated as $r = \sqrt{(11.5^2 + 40^2 + 20^2)} \text{ nm} = 46.17\text{nm}$, $k = 9 \times 10^9$, $q_1 q_2 = (1.6)^2 \times 10^{-38} \text{ C}^2$. So potential between them are:

$U_1 = \frac{23.04 \times 10^{-29}}{44.72 \times 10^{-9}} \text{ J} = 0.515 \times 10^{-20} \text{ Joule}$. By taking the distances between electron x and remaining electrons, table II can be constructed which shows total potential for polarization +1 of configuration 1 is $U_{T+1} = U_{T+1x} + U_{T+1y} = 6.5 \times 10^{-20} + 4.42 \times 10^{-20} \text{ Joule} = 10.92 \times 10^{-20} \text{ Joule}$. Similarly the total potential for polarization -1 of configuration 1 is $U_{T-1} = U_{T-1x} + U_{T-1y} = 4.70 \times 10^{-20} + 6.22 \times 10^{-20} \text{ Joule} = 10.92 \times 10^{-20} \text{ Joule}$. Hence configuration 1 has potential of $21.84 \times 10^{-20} \text{ Joule}$.

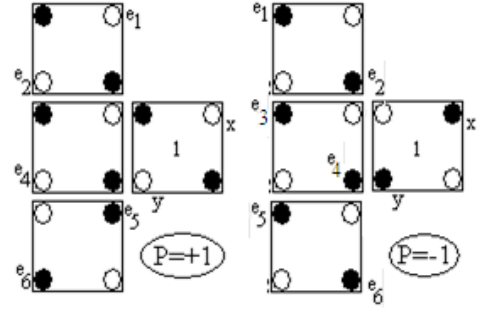


Figure 6. Configuration 2 of layered T gate (four cell structure)

Total potential for polarization +1 of configuration 2 mapped with figure 5 is $U_{T+1} = U_{T+1x} + U_{T+1y} = 12.76 \times 10^{-20} + 4.46 \times 10^{-20} \text{ Joule} = 17.22 \times 10^{-20} \text{ Joule}$ which can be obtained from table III. Similarly total potential for polarization -1 of configuration 2 is $U_{T-1} = U_{T-1x} + U_{T-1y} = 4.13 \times 10^{-20} + 6.75 \times 10^{-20} \text{ Joule} = 10.883 \times 10^{-20} \text{ Joule}$. Hence configuration 2 has potential of $28.103 \times 10^{-20} \text{ Joule}$. Though both of the configurations cause the same results, their potential reflections are different. Calculation of configuration 1 projects lower potential value than the potential of configuration 2. Hence this work uses configuration 1 Layered T gate for optimized design use.

C. Potential Stability -Layered T with Inverter

Layered T technique presents inverted NAND gate or equivalently AND gate by means of one layered T gate positioned with one corner cell. Research in inverter section of QCA experiences the progression from seven cell state of the art inverter [13] to two cell inverter [14]. The following figure 5 shows five cell layered T gate. To invert the output, another diagonally positioned cell requires, which has C and D

electrons as shown in the figure 7a). The distance between electron A and electron C of figure 7a), is $\sqrt{(11.5^2 + 60^2 + 20^2)} \text{ nm} = 64.28 \text{ nm}$ whereas $\sqrt{(11.5^2 + 78^2 + 38^2)} \text{ nm} = 87.52 \text{ nm}$ distance between A and point D requires another clock zone to evaluate output data at the same cell. It necessitates the requirement of two clocking cycle because this effort verifies its result in 75nm radius of effect. The distance between point A and point D exceeds the required distance.

But the same distances in figure 7b) shows the reduction of distance between electrons A & D up to $\sqrt{(11.5^2 + 58^2 + 20^2)} = 70.28 \text{ nm}$. That indicates the gate with the corner cell is included within the same clock zone. This analysis forces the author to use five cell corner layered T gate so as to avoid unnecessary delay in digital circuits.

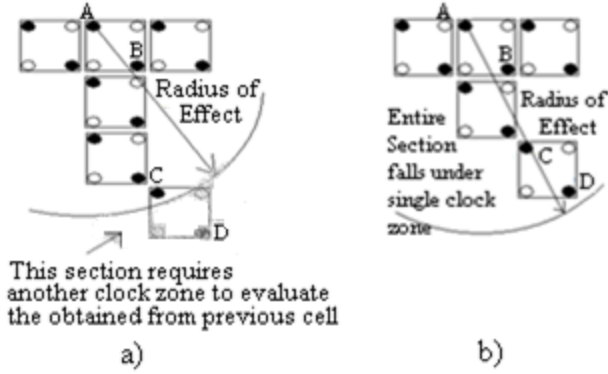


Figure 7. Layered T gate with one corner cell to implement Layered T-inverter

Scalability of layered T gate methodology mainly bests in the design of one bit full adder if aforementioned potentially

optimized gates are used in the designing of arithmetic logic unit.

IV. VERIFICATION OF LAYERED T-GATE IN FULL ADDER

One-bit full adder works as basic primitive for digital logic units. One bit full adder of [3] needs five majority gates which are further improved in the design proposed in [16] and [17]. If area required for fabrication is minimized then n-bit full adder can be further optimized making design smaller and effective. Processor architectures of recent trends and its high packing density emphasizes on “below 45nm” technology [18]. The proposed Layered T-gate reduction technique is used to generate the schematic of 1-bit full adder which needs only 85988.96nm² as in figure 8 and verified in QCADesigner [19]. The expression of 1-bit full adder is expressed as follows:

$$\text{SUM} = A \oplus B \oplus C \quad (4)$$

$$\text{COut} = AB + BC + CA \quad (5)$$

The proposed universal NAND/NOR based approach implements full adder by means of Ex-OR gates. Two successive Ex-OR gates along with one five cell AND gate are connected to get the sum and carry output. Any schematic (that means number of logic gates and their interconnections) promises the best result if the input and output ports are placed in the borders of the circuit and as a consequence additional wires and cross-wires do not require connecting the block with other logic blocks. If any ports are specified in middle of the circuits then additional wires increase the circuit area and operational cost of the circuit.

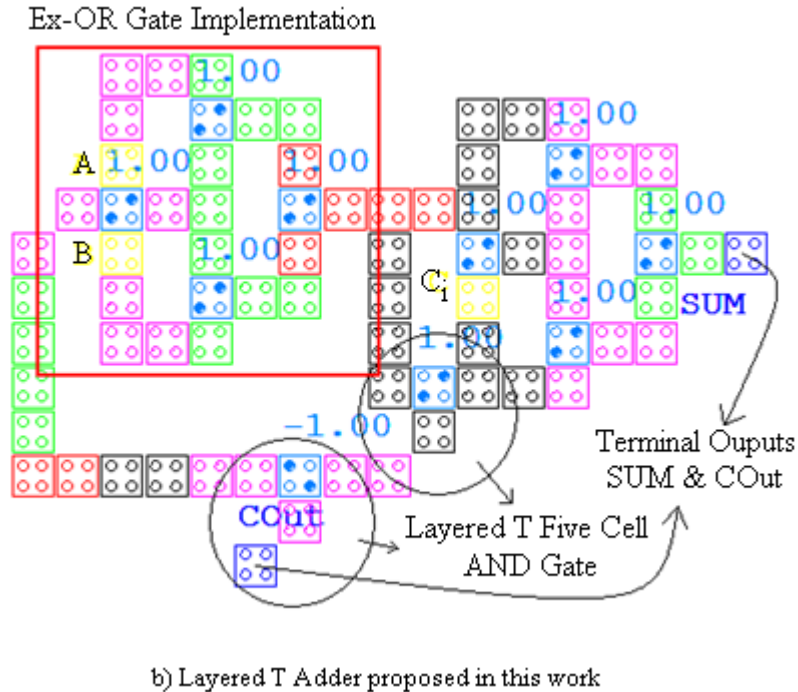


Figure 8. Layout of 1 bit full-adder using proposed Layered T reduction technique

Table III Potential Calculation of configuration 2 considering figure 6

Potential for P=+1 of configuration 2						Potential for P=-1 of configuration 2					
Sl. No.	Distance from electron x	Energy (in J)	Sl. No.	Distance from electron y	Energy (in J)	Sl. No.	Distance from electron x	Energy (in J)	Sl. No.	Distance from electron y	Energy (in J)
1.	to $e_1=28.28\text{nm}$	0.816×10^{-20}	7.	to $e_1=53.74\text{nm}$	0.429×10^{-20}	1.	to $e_1=42.94\text{nm}$	0.537×10^{-20}	7.	to $e_1=42.94\text{nm}$	0.537×10^{-20}
2.	to $e_2=2.82\text{nm}$	8.18×10^{-20}	8.	to $e_2=28.28\text{nm}$	0.816×10^{-20}	2.	to $e_2=20.09\text{nm}$	1.148×10^{-20}	8.	to $e_2=20.091\text{nm}$	1.148×10^{-20}
3.	to $e_3=23.07\text{nm}$	1×10^{-20}	9.	to $e_3=43.59\text{nm}$	0.52×10^{-20}	3.	to $e_3=39.702\text{nm}$	0.581×10^{-20}	9.	to $e_3=29.26\text{nm}$	0.78×10^{-20}
4.	to $e_4=21.45\text{nm}$	1.076×10^{-20}	10.	to $e_4=23.07\text{nm}$	1×10^{-20}	4.	to $e_4=29.261\text{nm}$	0.789×10^{-20}	10.	to $e_4=11.67\text{nm}$	1.97×10^{-20}
5.	to $e_5=20.09\text{nm}$	1.148×10^{-20}	11.	to $e_5=20.09\text{nm}$	1.14×10^{-20}	5.	to $e_5=42.941\text{nm}$	0.537×10^{-20}	11.	to $e_5=20.09\text{nm}$	1.14×10^{-20}
6.	to $e_6=42.94\text{nm}$	0.537×10^{-20}	12.	to $e_6=42.94\text{nm}$	0.53×10^{-20}	6.	to $e_6=42.94\text{nm}$	0.537×10^{-20}	12.	to $e_6=20.09\text{nm}$	1.14×10^{-20}
U_{T+1x}		12.76×10^{-20}	U_{T+1y}		4.46×10^{-20}	U_{T-1x}		4.131×10^{-20}	U_{T-1y}		6.75×10^{-20}

Moreover this work considers a cell with a fixed polarization of either +1 or -1, is equivalent to a state where two of four dots are filled with electrons. This is a stable state [3] [4] so it is possible to obtain these kinds of cells. The full adder proposed in this work provides an optimistic and realistic area optimization as the inputs A & B and output SUM & Carry are specified in the border of the circuit shown in figure 8. Full adders with efficient comparators [15] [20] can be further used for efficient processor designs in terms of delay and cross wires.

V. RESULTS & DISCUSSION

NAND-NOR based Layered T gate fulfills the necessity of universal logic in QCA circuit designing. Four and five cell configuration of layered T gate has shown the same results in the output but in terms of stability, five cell layered T gate wins the race. Decreased value of 28.103×10^{-20} Joule to 21.89×10^{-20} Joule observes sharp 22.27% improvement in energy calculation. Five cell Inverted Layered T Gate with one

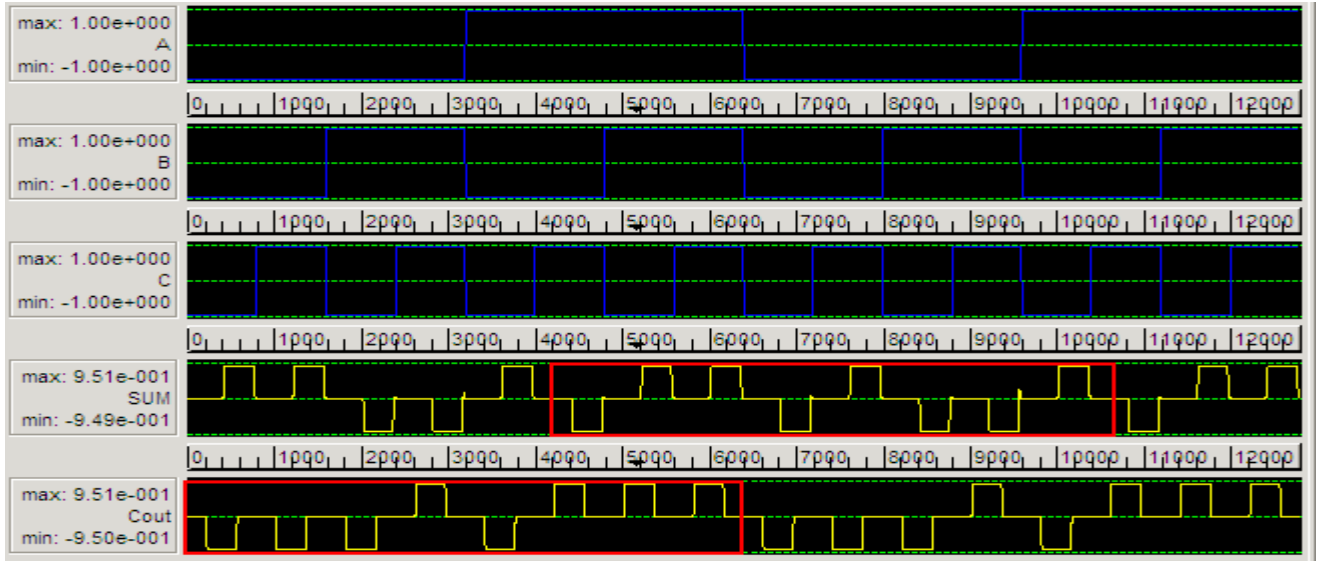


Figure 9. Output of the one-bit Layered T full adder simulated in QCADesigner [19]

A. Output Verification

The bit patterns of SUM and COut in figure 9 are observed as 01101001 & 00010111 which resemble the output graph of full adder. Bold portion of the graph shows the similarity with full adder graph in bistable approximation methodology. The Table III shows the comparison of one-bit full adders with the previous designs in terms of number of cell required and area.

corner cell requires only one clock zone to evaluate the data. Compared to six cells inverter configuration, superiority of five cell configuration can be observed. Proposed layered T Full adder requires 76 cells and 85988.96nm^2 area to get SUM and CARRY outputs. Previously designed adder, proposed by H. Cho et al [23], requires least number of cells whereas design proposed in this work needs 76 cells and $0.08\mu\text{m}^2$ area. Layered T Full adder gives the best feedback in terms of two key

parameters, cell count and area requirements for fabrication [4].

Table III Comparison of Layered T Full Adder with previous full adder

Sl. No.	Design Proposed by	Cell Count	Area
1	P.D.Tougaw et al [3]	192	0.2 μm^2
2	Wei Wang et al[16]	148	0.16 μm^2
3	M.B.Tahoori et al[21]	145	0.16 μm^2
4	R.Zhang et al in 2004 [17]	145	0.17 μm^2
5	V.C.Teja et al in 2008 [22]	124	>0.1 μm^2
6	H.Cho et al. in 2009 [23]	80	0.0923 μm^2
7	B.Sen et al in 2010 [6]	208	>0.2 μm^2
8	Proposed LT Full Adder	76	0.0859 μm^2

VI. CONCLUSION

Proposed five cell layered T gate and inverter with corner cell are potentially optimized and scalable to standard designs. Taking all the statistical analysis with simulation results, this work strongly proposes layered T Full adder which gives superior performance with respect to the previously designed adder and a methodology which implements universal logic.

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